POWERING THE NATION: CAN YOU CONTROL A NUCLEAR REACTOR?

An Introduction to Reactor Safety & Dynamics

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THE BIG QUESTION: POWERING SINGAPORE'S FUTURE

Singapore faces a major challenge known as the **Energy Trilemma**:

Secure Can we guarantee a stable supply that never fails?

Affordable Can we keep electricity costs low for everyone?

Sustainable Can we generate power without contributing to climate change?

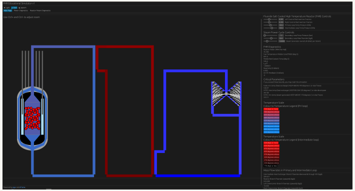
A Potential Solution?

Advanced nuclear reactors are being considered globally as a high-tech solution that could potentially address all three points. But are they safe?

YOUR MISSION, SHOULD YOU CHOOSE TO ACCEPT IT

Today, you are not just students. You are reactor operators at the controls of a next-generation Fluoride-salt-cooled High-temperature Reactor (FHR).

The Situation: The grid operator requests an immediate increase in power output to meet a sudden demand spike.



Your Objective

Increase reactor thermal power from **30 MWth** to **60 MWth** and stabilize it there.

OPERATIONAL CONSTRAINTS & SAFETY LIMITS

Nuclear reactors are powerful systems that require precise control. We cannot just ”floor the accelerator.”

The Rules of Engagement:

- You have full control over the reactor’s control rods (reactivity).
- You must aim for exactly **60 MWth**.

SAFETY LIMIT (FAILURE CONDITION)

If reactor power exceeds **1000 MWth** at any point during the maneuver, you have violated the reactor’s safety limits and failed the mission.

YOUR INTERFACE: THE FHR CONTROL PANEL

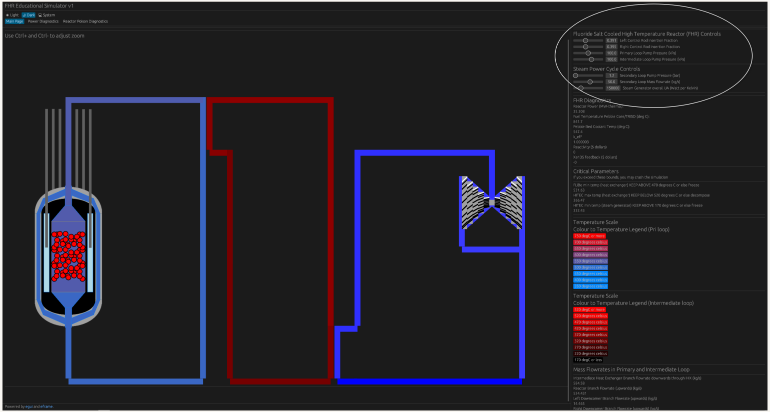


Figure: *Current State: Stable at 30 MWth*

Volunteers Needed!

Who has the steady hand to bring us to 60 MWth without exceeding 1000 MWth?

For the rest of the class (Observers): Watch closely. Don't just look at the power level.

- How does the reactor respond immediately after they move the controls?
- What happens to the power when they *stop* moving the controls?
- Does the reactor feel "heavy" (slow to move) or "twitchy" (fast to move)?

DEBRIEF: INTUITION VS. REALITY

It wasn’t as easy as turning up a volume knob on a radio.

Discussion Questions:

1. Did anyone overshoot the 60 MWth target? Why do you think that happened?
2. Did you notice the power continuing to change even after the controls were put back to neutral?
3. If you added too much reactivity, did it feel like the reactor was ”running away” from you?

Why does the reactor have ”momentum”? Why doesn’t it stop instantly?

To answer this, we need to understand what makes nuclear engineering a unique challenge.

IS IT REALLY THAT SIMPLE?

On paper, the concept of a nuclear reactor sounds straightforward:

> *"Neutrons split uranium to heat something that then boils water to spin a turbine."*

But you just experienced it for yourselves. It wasn't that simple.

- The power had "momentum."
- It was easy to overshoot the target.
- It didn't respond like a car engine or a volume knob.

IS IT REALLY THAT SIMPLE?

The Core Question

If the basic idea is so easy, why is controlling it so tricky?
The gap between that simple sentence and a safe, working power plant is precisely why nuclear science and engineering is such a deep and fascinating field.

THE JOURNEY FROM PAPER TO POWER PLANT

To use nuclear energy to help solve Singapore's energy trilemma, we must bridge that gap.

How do we get from a paper design to a safe, working plant that can operate for 60+ years?

It's a grand challenge with many layers:

- **Politics & Public Policy:** How do we regulate and govern this technology?
- **Economics & Finance:** How do we fund and build such a large project?
- **Social Acceptance:** How do we ensure public trust and communicate effectively?
- **Technical & Engineering Safety:** How do we guarantee the plant will be safe under all possible conditions?

THE JOURNEY FROM PAPER TO POWER PLANT

Our Focus Today: Technical Safety

My work, and what we'll explore next, lives in that last box. This is the world of **Reactor Safety**, where we must understand every physical process inside the core.